nature portfolio

Corresponding author(s): Liang Qu

Last updated by author(s): May 6, 2026

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Flow cytometry data were acquired using the Cytek Aurora (Cytek). Western blot and gel data were acquired using the Tanon 5200 Multi (Tanon). LC-MS/MS data were acquired using the Orbitrap Eclipse (Thermo Fisher Scientific). RNA-seq data were acquired using the NovaSeq 6000 (Illumina). qPCR data were acquired using the QuantStudio 6 Flex (Thermo Fisher Scientific). ELISA data were acquired using the Varioskan LUX (Thermo Fisher Scientific). Animal imaging data were acquired using the SII in vivo imaging system (Bruker). VLP features were acquired using the Transmission Electron Microscope HT7800 (Hitachi).
Data analysis	FlowJo (v10) was used to analyze flow cytometry data. ImageJ (1.53c) was used to analyze images. GraphPad Prism 9 (GraphPad) was used for analysis of phenotypic assays and for graph production. RNA-seq read quality was assessed using Trimmomatic v0.39. Filtered reads were aligned to the human or mouse reference genome (hg38/mm39) using HISAT2 v2.2.1. Gene-level counts were generated with FeatureCounts v2.0.6. Differential expression analysis was performed using Bioconductor package DESeq2 v1.46.0. Immune-related genes were defined based on the ImmPort database. Gene Set Enrichment Analysis (GSEA) and Gene Ontology (GO) enrichment analysis were performed using clusterProfiler v4.14.6. Data visualization was carried out using ComplexHeatmap v2.22.0 and ggplot2 v3.5.2. Microsoft Excel v16 was used for statistical analysis. Adobe Illustrator (29.0.1) was used to create all original materials and graphical elements in this manuscript.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

All data supporting the findings of this study are available within the article, the Supplementary Information files or Source Data files. The accession code for the transcriptome-wide RNA-sequencing data in this study is GSE303451 in the NCBI Gene Expression Omnibus (GEO) database. Supplementary Information files or Source Data files are provided with this paper.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	Peripheral blood mononuclear cells used in the study were obtained from one de-identified male donor.
Reporting on race, ethnicity, or other socially relevant groupings	Race, ethnicity and other socially relevant groupings were not specifically considered in this study.
Population characteristics	This study used human primary T cells isolated from the peripheral blood of one de-identified male donor aged 20–40 years.
Recruitment	The donor was randomly recruited and written informed consent was obtained.
Ethics oversight	This study was approved by the Ethics Committee of Fudan University.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Monitoring of tumour growth was performed using n ≥ 5 tumour?bearing mice per condition. Immunogenicity test was performed using n ≥ 4 BALB/c mice per condition. IPGTT experiment was performed using n ≥ 4 C57BL/6J mice per condition. CircRNA-GLP-1 therapy was performed using n = 4 DIO mice per condition. hm5C-circRNA-MOG therapy was performed using n ≥ 4 C57BL/6J mice per condition. Bioluminescence imaging was performed using n = 3 BALB/c mice per condition. All cell samples were assessed in at least three biological replicates to enable statistical analysis. All remaining sample sizes are provided in the captions of the respective figures. The sample size was decided based on previous publications with similar experiments.
Data exclusions	No data were excluded.
Replication	All replication attempts were successful. All experiments were performed with at least two independent replicates.
Randomization	Mice were acclimated at the animal facility for at least 7 days before performing any experiments. Allocation of mice was randomized and experimental and control animals were treated equally. For other experiments, samples were randomly allocated to experimental groups.
Blinding	Blinding was not required for this study, given that results were based purely on objective assessments.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used: Fixable Viability Dye BioLegend 423102; CD45 APC I3/2.3 BioLegend 147708; CD45 PerCP 30-F11 BioLegend 103130; CD3 PE-Cy5 17A2 BioLegend 100274; NK1.1 AF700 S17016D BioLegend 156512; CD11b PerCP M1/70 BioLegend 101229; F4/80 BV421 BM8 BioLegend 123132; H-2Kb bound to SIINFEKL PE 25-D1.16 BioLegend 141603; CD11c BV650 N418 BioLegend 117339; CD3 PE-Cy5 17A2 BioLegend 100274; CD4 APC-Cy7 GK1.5 BioLegend 100414; CD4 PE-Cy7 RM4-4 BioLegend 116016; Foxp3 Pacific blue MF-14 BioLegend 126410; PD-1 BV711 29F.1A12 BioLegend 135231; CD8 APC-Cy7 53-6.7 BioLegend 100713; CD69 FITC H1.2F3 BioLegend 104505; IL-17A BV605 TC11-18H10.1 BioLegend 506927; IFN-γ PE XMG1.2 BioLegend 505808; Anti-DYKDDDDK Tag PE/Dazzle 594 L5 BioLegend 637330; Anti Flag M2 Mouse Monoclonal Antibody Merck #F1804; Anti β-Tubulin Mouse Monoclonal Antibody 3G7 CoWin Biosciences #CW0098M; Poxidase AffiniPure Goat Anti-Mouse IgG (H+L) secondary antibody Jackson ImmunoResearch #115035003.

Validation: All antibodies were bought from vendors. All antibodies were validated by the suppliers and validation information is available on the manufacturers' websites.

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s): HEK293T (ATCC, CRL-3216) was maintained in Liang Qu's laboratory at Fudan University. B16F10-OVA (#ZQ1169), HeLa (#ZQ0068) and A549 (#ZQ0003) cells were purchased from Shanghai Zhong Qiao Xin Zhou Biotechnology Co., Ltd. and MC38-OVA cell line (#IMO-019) was purchased from Xiamen Immocell Biotechnology Co., Ltd.

Authentication: All cell lines were authenticated by morphology.

Mycoplasma contamination: All cell lines were confirmed to be negative for mycoplasma contamination.

Commonly misidentified lines (See ICLAC register): No commonly misidentified cell lines were employed in this study.

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals: This study employed female C57BL/6J mice (6–8 weeks), female BALB/c mice (6–8 weeks), male diet-induced obese mice (15–16 weeks), and female CAG-loxp-zsGreen-stop-loxp-tdTomato-PolyA Cas9-KI reporter BALB/c mice (6–8 weeks). All mice were housed and maintained under specific pathogen-free (SPF) conditions in the Laboratory Animal Center of Fudan University. Mice were given free access to food and water, and kept under a 12-h light–dark cycle between 21-24°C and 50-60% humidity. Especially, DIO mice were fed a 60% kcal high-fat diet.

Wild animals: No wild animals were involved in this study.

Reporting on sex: Sex-based analyses were not performed in this study

Field-collected samples: No field-collected samples were used in this study.

Ethics oversight: All animal experiments were approved by the Committee on the Ethics of Animal Experiments of Fudan University (20240229-101) and undertaken in accordance with the National Institute of Health Guide for Care and Use of Laboratory Animals.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks: The study did not involve plants.

Novel plant genotypes: -

Authentication: -

Flow Cytometry

Plots

Confirm that:

☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☒ All plots are contour plots with outliers or pseudocolor plots.

☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation: For the analysis of spleen and lymph nodes, these tissues were removed from mice and ground into single-cell suspensions. For the analysis of tumor infiltrating immune cells, tumors were removed from tumor-bearing mice and cut into fragments. Tumor samples were digested in RPMI-1640 medium containing 1% fetal bovine serum, 1 mg/mL collagenase type IV (Sigma, #C5138) and 0.5 mg/mL DNase I (Roche, #10104159001) at 37°C for 30 min and then tumor fragments were incubated on a shaker at 37°C for 30 min. Following adequate digestion, tumor samples were put on ice in order to terminate the reaction and then were ground into single-cell suspensions. The cells were collected at the interphase between 70% and 42% percoll (Cytiva, #GE17-0891-02) and washed with FACS buffer.
For the analysis of brain and spinal cord infiltrating immune cells, brains and spinal cords were removed from EAE mice and cut into fragments. Samples were digested in RPMI-1640 medium containing 1% fetal bovine serum, 0.5 mg/mL papain (Sangon, #A003118) and 0.25 mg/mL DNase I (Roche, #10104159001) by shaking at 37°C for 20 min. Then the samples were ground into single-cell suspensions and the lymphocytes were extracted by percoll and washed with FACS buffer.

Instrument: Cytek Aurora (Cytek)

Software: FlowJo v10

Cell population abundance: The target cell populations were sufficiently abundant for analysis.

Gating strategy: Representative schemes for gating and threshold setting are shown in the Supplementary figure. Unedited cells were employed as negative control for gating the positive signal.
For OVA-specific DCs and macrophages, the cells were first gated based on FSCA/SSCA and FSCA/FSCH to select for live single cells. The cells were gated based on CD45/ Live/Dead to select for live lymphocytes. The lymphocytes were gated based on NK1.1/ CD3 to select for NK1.1- CD3- lymphocytes. The NK1.1- CD3- lymphocytes were gated based on SSC/ CD11b to select for CD11b+ lymphocytes. The CD11b+ lymphocytes were gated based on CD11b/ F4/80 to select for macrophages or CD11b/ CD11c to select for DCs. The macrophages were gated based on H-2Kb/SIINFEKL/ F4/80 to select for OVA-specific macrophages and The DCs were gated based on H-2Kb/SIINFEKL/ CD11c to select for OVA-specific DCs.
For Tregs, the cells were first gated based on FSCA/SSCA and FSCA/FSCH to select for live single cells. The cells were gated based on FSC/ Live/Dead to select for live cells. The live cells were gated based on FSC/ CD45 to select for lymphocytes. The lymphocytes were gated based on CD3/ CD4 to select for CD4+ T cells. The CD4+ T cells were gated based on Foxp3/ CD4 to select for Tregs.
For CD4+ PD-1+ T cells, the cells were first gated based on FSCA/SSCA and FSCA/FSCH to select for live single cells. The cells were gated based on FSC/ Live/Dead to select for live cells. The live cells were gated based on CD3/ CD45 to select for T cells. The T cells were gated based on CD3/ CD4 to select for CD4+ T cells. The CD4+ T cells were gated based on PD-1+/ CD4 to select for CD4+ PD-1+ T cells.
For CD69+ CD4+ T cells, the cells were first gated based on FSCA/SSCA and FSCA/FSCH to select for live single cells. The cells were gated based on FSC/ Live/Dead to select for live cells. The live cells were gated based on SSC/ CD45 to select for lymphocytes. The lymphocytes were gated based on CD8/ CD4 to select for CD4+ T cells. The CD4+ T cells were gated based on CD69/ CD4 to select for CD69+ CD4+ T cells.
For IL17-A+ CD4+ T cells and IL17-A+ IFN-γ+ CD4+ T cells, the cells were first gated based on FSCA/SSCA and FSCA/FSCH to select for live single cells. The cells were gated based on FSC/ Live/Dead to select for live cells. The live cells were gated based on FSC/ CD45 to select for lymphocytes. The lymphocytes were gated based on CD3/ CD45 to select for T cells. The T cells were gated based on CD8/ CD4 to select for CD4+ T cells. The CD4+ T cells were gated based on IL17-A/ CD4 to select for

IL17-A+ CD4+ T cells or IL17-A/ IFN-γ to select for IL17-A+ IFN-γ+ CD4+ T cells.

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.